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Cloud Auto-Scaling and load balancing during seasonal sales.

Companies often experience a sudden increase in website traffic during festivals or promotional events. Cloud auto scaling automatically increases the number of servers when demand rises and reduces them when traffic becomes normal. Load balancing distributes user requests across multiple servers so that no single server becomes overloaded. This ensures smooth website performance even during heavy traffic.

Key points:

- Auto scaling adds or removes resources automatically.
- Load balancing shares incoming traffic evenly.
- Reduces website crashes and downtime.
- Improves response time and customer satisfaction.

Conclusion:- cloud auto-scaling and load balancing help businesses maintain reliable services, reduce downtime and provide a better user experience.

2 cloud storage for Security backup:

Introduction: organization needs secure and reliable storage for important backup data. cloud storage offers a flexible solution with high availability and security.

Explanation:- cloud providers store multiple copies of data across different locations, ensuring durability and redundancy. encryption protects data while storing and transferring it. access control allows only authorized users to access backup files compared to traditional storage, cloud storage eliminates the need for expensive hardware and maintenance, reducing costs.

Example:- if company's office is damaged by a natural disaster. all important files stored in the cloud can be restored quickly allowing business operations to continue.

Summary: cloud storage provides secure, durable and cost effective backup while supporting disaster recovery.

Global Application deployment using CDN

What is a CDN?

A content delivery network (CDN) automatically delivers it from the nearest server instead of giving data from a single central server. A CDN stores copies of frequently accessed content such as images, videos, application files, and web pages on servers.

Advantages of CDNs:-

- Reduces loading time.
- Minimizes network latency.
- Handles large amounts of user traffic.
- Reduces the workload on the main server.

How cloud providers supports global availability

Cloud providers have data centres in many countries. Applications can be deployed in multiple regions so that if one data centre fails, another one continues providing services. Load balancing and automatic fail over ensure continuous availability.

Conclusion:-

Content delivery network, edge servers and globally distributed cloud infrastructure improve application speed, reduce latency and provide high availability. These technologies are essential for applications.

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Cloud Security and compliance for financial institutions.

why do financial institutions require strict cloud security?

Financial institutions store confidential customer information, account details, transaction records and payment info. any data breach can lead to financial loss and legal issues. therefore, strong security and regulatory compliance are essential while using cloud services.

why is encryption important?

Encryption converts readable information into an unreadable format. data remains encrypted while being stored in cloud servers and while travelling through the internet. even if attackers obtain the data they cannot understand it without the encryption key.

Example:

Consider a bank that provides online banking services. every customer transaction is being encrypted before storage.

Conclusion:-

cloud providers offer compliance support, encryption, logging monitoring and regional data storage to help financial institutions information

5. Containerization for application deployment.

Introduction:

Modern software applications require faster deployment, efficient resource utilization, and easy scalability. Containerization has become one of the most popular technologies in cloud computing because it allows developers to package applications along with all required dependencies into lightweight containers. These containers can run consistently across different environments.

Containers vs Virtual machines:-

<u>Feature</u>	<u>containers</u>	<u>virtual machines</u>
Operating sys	share it host operating sy.	Each vm has its own operating system.
Start up time	starts within seconds	takes several minutes.
Resource usage	use less cpu and memory	Requires more hardware resources.
Performance	Faster	comparatively slower.
portability.	highly portable.	Less portable.

Containerization provides lightweight, fast and portable application deployment, while Kubernetes simplifies management through automation, scaling and self healing capabilities.